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Intelligent charge scheduling and eco-routing mechanism for electric vehicles: A multi-objective heuristic approach

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ABSTRACT

Due to the rising pollution and greenhouse gas emissions resulting from fossil fuel-based transportation systems, researchers and policymakers are pushing for Electric Vehicle (EV) that is envisaged as an efficient, eco-friendly alternative. However, due to their limited range and battery capacity, EVs need frequent charging, which is time-consuming and available at specific locations. Therefore, proper charge scheduling and route management of EVs is essential and significant. This paper addresses this problem by proposing an intelligent heuristic mechanism that ensures that the EVs are always routed through a path that minimizes the energy consumption and the total time to travel. We formulate it as a multi-objective optimization problem considering real-world specifications and constraints and propose a graph-based multi-objective heuristic algorithm (MoHA) to obtain the desired solutions quickly. Further, multiple variants of the proposed algorithm are proposed, and comparative analysis is performed on practical datasets. The proposed algorithm is evaluated based on some of the well-known performance metrics for multi-objective approaches. The results obtained show that the *energy-aware*-MoHA variant produced 32.39% better results in minimizing energy consumption, and *time-aware*-MoHA performed better in optimizing average time requirements by 24.32%. Moreover, the initial ordering of the EVs has significant importance on the proposed algorithm's overall performance.

1. Introduction

The automobile industry is one of the fastest-growing industries in today's manufacturing sector, which is predominantly driven by consumer preference-oriented intense global competition and innovation. However, the growing demand for and continuous increase in the number of vehicles has resulted in depleting non-renewable resources and the deterioration of the earth's atmosphere. Along with global population and urbanization, the current transportation system has been one of the significant contributors to the number of Greenhouse Gases (GHG) emitted that has brought severe environmental concerns and alarming pollution levels (Ramachandra, Aithal, & Sreejith, 2015). According to the Annual Energy Outlook 2018, published by the U.S. Energy Information Administration, approximately 28% of energy consumption in the U.S. has been due to transportation in 2017 (Annual Energy Outlook, 2020). Similar statistics have also been obtained by the European Environment Agency that reported the transportation sector to result in 25.8% of total EU-28 GHG emissions (EEA, 2017).

This has consequently led governments, environmental agencies, and automobile manufacturers to develop alternative modes of transportation. Electric vehicle (EV) is considered to be a promising and desirable alternative to the fossil fuel-based transportation system that has been envisioned as an efficient mechanism to address the global concern of GHG emissions and reduce the heavy reliance on fossil fuels for the current transportation systems (Wang, Liu, Du, & Kong, 2016). Compared to conventional vehicles, EVs introduce numerous benefits such as low noise, low carbon, ozone-precursor emissions, high energy efficiency, etc. However, despite the various advantages, widespread adoption of EVs is still obscured due to the factors such as battery life, charging time, availability of charging stations, the cost for charging, and safety (Lokesh & Min, 2017). EVs suffer from low engine capacity that makes the journey comparatively slower than traditional vehicles. The batteries have limited charge storage capacities, which reduce the range of distance covered in one go and require frequent charging that takes a significantly longer period of time than the traditional fuel refilling process, possessing significant challenge given the current charging infrastructure (Moghaddam, Ahmad, Habibi, & Phung, 2018).

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Acronyms

CS	Charging station
DBC	Decreasing battery capacity
DE	Decreasing energy consumption
EV	Electric vehicle
ER	Error ratio
GHG	Greenhouse gases
HVR	Hyper-volume ratio
GD	Generational distance
IBC	Increasing battery capacity
IE	Increasing energy consumption
IS	Increasing speed
MoHA	Multi objective heuristic algorithm
NDSP	Non-dominated shortest path
DS	Decreasing speed
SS	Schott's spacing metric

A significant amount of existing literature tried to address the aforementioned challenges, particularly the charging issue of EVs. Researchers have proposed energy-efficient routes and various charging profiles for the vehicles considering real-world challenges (Battarra, Gargiulo, Tremiterra, & Zucaro, 2018; Kin, Verlinde, & Macharis, 2017). Sayarshad, Mahmoodian, and Gao (2020) a Markov Decision Process-based approach to obtain the best routes for electric taxis that are assigned to consumers with elastic demands. An energy-efficient route is defined to be a path for an EV from the source to the destination that consumes the least amount of energy, also known as 'eco-routing' (Boriboonsomsin, Barth, Zhu, & Vu, 2012; Cela et al., 2014; Nie & Li, 2013). Boriboonsomsin et al. presented one such early work in Boriboonsomsin et al. (2012) where they proposed an eco-routing navigation system. Authors in Cela et al. (2014) proposed a real-time eco-routing framework that captures not only energy consumption but also on-board power splitting between batteries and super-capacitors. Le Rhun, Bonnans, De Nunzio, Leroy, and Martinon (2020) proposed an eco-routing mechanism considering the traffic conditions on the road. They utilized the A^* algorithm to solve the optimal pathfinding problem. Authors in Yang et al. (2017) presented an EV routing mechanism based on crowdsensing. Han, Han, and Sezaki (2010) used dynamic programming for optimal charge controlling of EVs along with vehicle-to-grid frequency regulation services. Authors in Gan, Topcu, and Low (2013) proposed a distributed optimal charge scheduling algorithm for EVs; however, their focus was to flatten out the grid load profile. A similar valley filling objective by EV scheduling was also targeted by Rivera, Goebel, and Jacobsen (2017). A distributed charging algorithm was also proposed by Floch, Belletti, and Moura (2016) where they explored the principle of duality, considering the perspective of both the EVs and the utility grid. Yi and Bauer (2018) presented a stochastic optimization model for routing EVs that incorporates uncertainties related to future charging demands.

Apart from eco-routing, a significant amount of research has been carried out on scheduling EVs' charging process at charging stations. Brady and O'Mahony (2016) proposed probabilistic modeling of daily travel and charging profiles for a set of EVs, which are then used to model the power demands. The overall objective of their study is to determine the schedules and charging patterns to have a better decision-making system for the EVs. Authors in Minett, Salomons, Daamen, van Arem, and Kuijpers (2011) proposed graph-based modeling of the EV charge scheduling problem and proposed a routing method based on Dijkstra's shortest path algorithm. However, the proposed routing method is more suited for the traditional source-to-destination routing problems. Pourazarm, Cassandras, and Malikopoulos (2014) suggested a pathfinding algorithm based on dynamic programming by considering graph-based modeling of the problem. Their overall objective has been to reduce the total time elapsed for a tour that includes charging time as well as traveling time. Similarly, Qin and Zhang (2011) targeted minimizing waiting time with the help of a distributed charge scheduling scheme. Authors in Tong (2019) developed a mechanism to obtain the driving cycle for public electric buses in a city network, specifically for the Supercapacitor bus deployed in Hong Kong. Yao, Liu, Lu, and Yang (2020) also addressed a similar scheduling problem of electric buses to minimize annual total scheduling costs, including the purchase costs of the vehicles and chargers, the operating expenses of deadheading and timetabled trips, etc. Mukherjee and Gupta have presented a comprehensive summary of EV charge scheduling in Mukherjee and Gupta (2015).

Most of the aforementioned existing literature has used either energy-optimized routing or time-efficient routing to address the limited battery capacity and charge scheduling problem. However, only energy-efficient routes or time-efficient routes may not always be desirable, particularly for real-world scenarios. For example, two separate paths may exist from a particular source to a destination where one of them is energy efficient with higher time requirements due to traffic congestion. The other one is time efficient due to less congestion but consumes more energy because of the road conditions. Thus, in practice, there exists a trade-off between energy-efficient and time-efficient routes. In this work, our objective is to address this multi-objective optimization problem and obtain trade-off solutions efficiently. Some previous works towards the same research direction were carried out in Houshmand and Cassandras (2018), Nunzio, Thibault, and Sciarretta (2017), Sun and Zhou (2016). Houshmand and Cassandras (2018) presented a combined vehicle routing and Power-train Control mechanism with the target to calculate the optimal energy route as well as the optimal power-train control strategy. Sun and Zhou (2016) addressed this problem by proposing an optimal cost algorithm to obtain a trade-off between traveling cost and time consumed, focusing on plug-in EVs. Nunzio et al. (2017) presented a graph-based bi-objective solution of eco-routing that minimizes both energy consumption and time. However, they mostly solved the problem by reducing the two objectives into a single objective problem using the weighted sum method. However, such a solution lacks clarity as obtaining proper weights for objectives is challenging. Moreover, the models presented in the majority of the existing literature assume that the EVs always travel on flat surfaces, which is not practical. For example, the energy consumption and time requirement will be different while traveling up a hill than getting down to a lower altitude point. Therefore, the applicability of the existing works on uneven surfaces, such as hill stations, is questionable.

Motivated by the above findings and the research gap in the literature, in this paper, we propose a multi-objective optimization framework for charge scheduling and eco-routing of EVs that minimizes both the average touring time and the average energy consumption. The touring time includes time to travel on roads, waiting time at the Charging Station (CS), and charging time at the CSs. Although there have been several works that tried to solve similar problems (Le Rhun et al., 2020; Nunzio et al., 2017; Zhang, Luo, & Li, 2016), a limited number of works exist that focus on the multi-objective optimization problem discussed in this work. Moreover, the existing solutions' poor efficiency makes these algorithms difficult to scale up in real life. Compared to the current literature, the novelty and unique contributions of this work can be listed as follows:

1 We discuss the multi-objective optimization problem of eco-routing and charge scheduling of EVs. We present a detailed mathematical formulation for the problem whereby the traffic network is modeled as a graph. The nodes in the graph represent junctions in the city and charging stations, which can be at various altitudes, represented by their respective elevation parameters. The graph's edges indicate

Table 1
Nomenclature

Notations	Descriptions
m	Number of charging stations
N	The set of n EVs
$G(V, E)$	The traffic network
$e_{pq} \in E$	The path from node v_p to v_q
D_{pq}	Distance from v_p to v_q
$\mathcal{E}_{-}$	Elevation at v_p
v_0^i	Source node for EV i
v_d^i	Destination node for EV i
T_0^i	Starting time for EV i
$\mathcal{B}^i$	Total Battery capacity of EV i
$\mathcal{R}^i$	Total range of EV i when fully charged
$\mathcal{S}_{\text{pq}}^i$	Input speed of EV i on e_{pq}
$\mathcal{W}^i$	Average energy consumed by EV i
$\mathcal{B}_h^i$	Threshold battery capacity
$\mathcal{B}_t^i$	Battery capacity of EV i at time t
$\mathcal{B}_{\rightarrow v_j}^i$	Battery capacity of EV i when entering the node v_j
$\mathcal{B}_{v_j \rightarrow}^i$	Battery capacity of EV i when leaving the node v_j
r_t^i	Running status of EV i at time t
c_t^{ij}	Charging status of EV i at CS j at time t
w_t^{ij}	Waiting status of EV i at CS j at time t

possible paths in the network with weights corresponding to the distance between the two connecting nodes.

2 We propose an intelligent Multi Objective Heuristic Algorithm (MoHA), which is essentially a graph-based scheduling approach that utilizes the multi-objective A^* search algorithm to obtain the desired solutions efficiently. We present four variants of the MoHA, viz., *energy-aware*, *time-aware*, *random*, *weighted*, each of which follows different strategies to break ties among multiple potential non-dominated solutions.

3 Unlike the existing similar multi-objective optimization solutions that produce results after transforming the problem into a single objective, the proposed heuristic algorithm produces an approximated Pareto front. We evaluate the quality of the solutions produced by the proposed MoHA and its variants in comparison to the Pareto optimal solutions for up to 20 devices using the well-known performance metrics for multi-objective heuristic solutions. In particular, the tools used are Error Ratio (ER), Generational Distance (GD), Schott's Spacing Metric (SS), and Hyper-volume Ratio (HVR). These metrics measure the quality of the Pareto front obtained using MoHA with respect to the relative distance from the optimal Pareto front.

4 The proposed MoHA is also evaluated using practical datasets consisting of graphs generated from the real-world scenario. Comparative performance analysis has been performed among the four variants of MoHA for up to 10000 EVs. The results obtained show that the variants produce trade-off solutions to the multi-objective optimization problem quite efficiently. Through simulation studies, we also analyze the effect of the elevation parameter on the overall quality of the results obtained by MoHA.

The rest of the paper is structured as follows. Section 2 presents the system model and introduces the multi-objective optimization problem formally. In Section 3, we propose the multi-objective heuristic algorithm. Simulation results are presented and discussed in Section 4. Finally, some concluding remarks and future works are presented in Section 5.

2. System model and problem description

Consider a traffic network represented as a graph $G(V, E)$. The nodes in the graph G mainly represent the CSs (a total of m CSs), and the

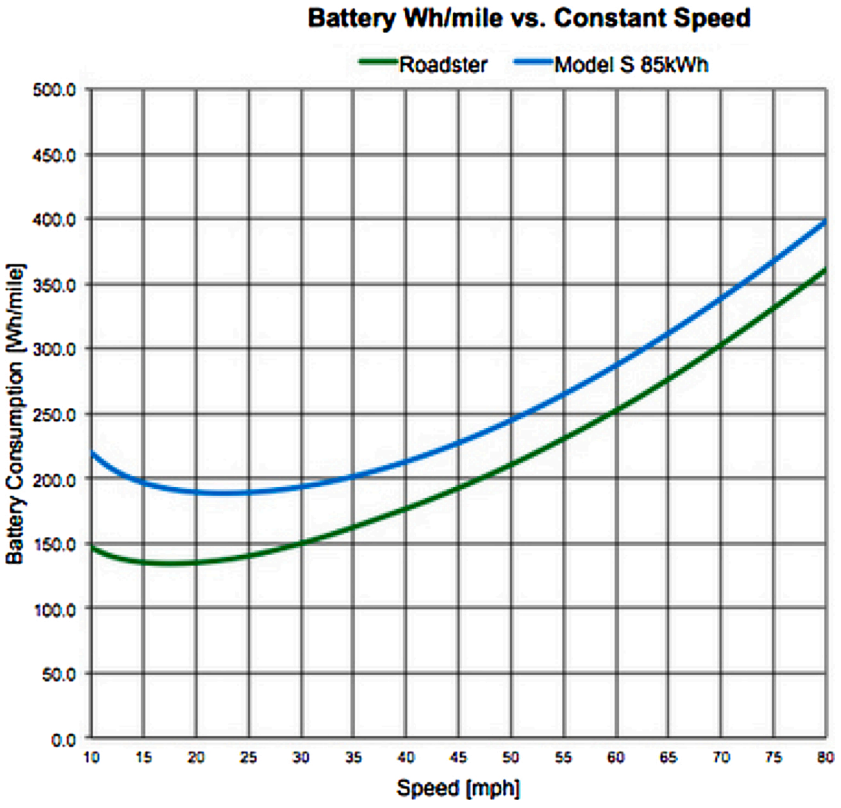


Fig. 1. Battery energy consumption per mile for Tesla Roadster and Tesla Model S 85 kWh (Musk & Straubel, 2012).

sources and destinations of the EVs. Connecting the nodes are the edges that represent possible paths among them. Each node $v_p \in V$ is characterized by an elevation parameter ($\mathcal{E}_p$), which is a measurement of their physical altitude. Each edge $e_{pq} \in E$ represents a direct path from v_p to v_q, and is assigned the weight D_{pq}, which represents the actual ground distance between the nodes v_p to v_q. The actual time required to travel along e_{pq} depends not only on the distance D_{pq} but also on the difference between the elevation parameters between v_p to v_q. For example, along the edge e_{pq}, if $\mathcal{E}_q > \mathcal{E}_p$ (indicating upward slope), the time required to travel is certainly would be more as compared to a condition where $\mathcal{E}_q = \mathcal{E}_p$. Furthermore, we consider each CSs to have a single charging port, although this can be extended for CSs with multiple charging ports by adding zero-weighted edges among the ports within the same CS.

Table 1

Let there be n EVs that need to travel from their sources to their respective destinations within the traffic network. Each of the EVs ($i \in N$, where $N = 1, 2, \ldots, n$) has individual attributes such as source ($v_0^i \in V$), destination ($v_d^i \in V$), starting time (T_0^i), battery capacity ($\mathcal{B}^i$), the total range ($\mathcal{R}^i$) in the unit distance when fully charged, and the energy consumption $\mathcal{W}^i$ per unit of time traveled. All these EV-specific parameters are input to the system and are assumed to take appropriate measurement units used in real-life. For simplicity, in this model, an EV i is considered to be charged non-preemptively till a threshold battery capacity $\mathcal{B}_h^i$. We assume that $\mathcal{B}_t^i$ represents the battery energy capacity at time t with $\mathcal{B}_0^i$ be the initial battery capacity of EV i at v_0^i, $\mathcal{B}_{\rightarrow v_j}^i$ be the battery capacity of EV i when entering the CS v_j, and $\mathcal{B}_{v_j \rightarrow}^i$ indicate the battery capacity with which EV i is leaving the CS v_j. Let us define a state variable $r_t^i \in \{0, 1\}$ to indicate the running status of EV i (1 if running, 0 otherwise) at time t. Similarly, we define the variable $c_t^{ij} \in \{0, 1\}$ to represent the charging status of EV i at CS v_j at time instant t, and $w_t^{ij} \in \{0, 1\}$ to indicate the waiting status of EV i at CS v_j at time t. Here we model time as a discrete entity. In between the source and its destination, an EV i is assumed to be either running, waiting, or charging. Therefore, we must have:

$$r_t^i + c_t^i + w_t^i = 1, \quad \forall t \tag{1}$$

where $c_t^i = \sum c_t^{ij}$ and $w_t^i = \sum w_t^{ij}$, $\forall j \in \{1, 2, \ldots, m\}$. Here, we consider time to be a discrete entity and therefore, we express battery energy capacity at any given time t as:

$$\mathcal{B}_t^i = \mathcal{B}_{t-1}^i - r_t^i \times \mathcal{B}_-^i + c_t^i \times \mathcal{B}_+^i \tag{2}$$

where $\mathcal{B}_+^i$ and $\mathcal{B}_-^i$ represent the amount of energy charged and discharged in the battery per unit time, respectively. In practice, the amount of energy charged depends on the charging port (fast charger, slow charger, etc.) and the charging power. In contrast, battery discharge may depend on multiple factors, including distance, the vehicle's speed, the slope of the road, wind direction, etc. (del Razo & Jacobsen, 2016). Fig. 1 shows an example of the variation in energy consumption with respect to speed (Musk & Straubel, 2012).

In this modeling, we consider both time (T_{pq}^i), and energy consumption ($\mathcal{B}_{pq}^i$) for the EV i along the edge e_{pq} to be dependent on the input speed of the vehicle $\mathcal{S}_{pq}^i$ on e_{pq}. In practice, there are several parameters, such as traffic conditions, the speed limit of the road, the slope of the road, etc., which might affect the energy consumption for the EV due to the higher power demand from the electric motor to maintain the same speed. Here we incorporate the slope of the path e_{pq} to be affecting the actual speed of the EV and, therefore, the effective speed ($\mathbb{S}_{pq}^i$) of EV i on e_{pq} is given as:

$$\mathbb{S}_{pq}^i = \frac{\mathcal{S}_{pq}^i}{1 + (\mathcal{E}_q - \mathcal{E}_p)} \tag{3}$$

where $\mathcal{E}_q$ and $\mathcal{E}_p$ are the elevation parameters of v_q and v_p. In the real-world scenario, the elevation of any particular node would be a numerical value usually measured in meters or kilometers. However, throughout this study, we assume that the elevation parameter of a node represents a scaled value of the true elevation value within a range of 0 to 1. With these information, we compute the time and battery energy consumed for EV i from v_p to v_q as follows (Vatanparvar & Faruque, 2017):

$$T_{pq}^i = \frac{D_{pq}}{\mathbb{S}_{pq}^i} \tag{4}$$

$$\mathcal{B}_{pq}^i = \mathcal{W}^i \cdot T_{pq}^i \tag{5}$$

It is important to note that the slope of the path changes the speed of the vehicle $\mathcal{S}_{pq}^i$ to $\mathbb{S}_{pq}^i$. Therefore, to maintain a constant effective speed on an uphill road, the input speed needs to be increased. Similarly, on downhill roads, higher effective speed can be achieved with lower input speeds. This eventually affects the actual time to travel from v_p to v_q, which, in turn, affects the battery energy consumption. Thus, on a path with upward inclination, an EV will consume more energy but travel less distance, in contrast to what it would have traveled had it been a non-inclined path. Consequently, the scenario is the opposite of a path with a downward inclination. For paths with no inclination, $\mathbb{S}_{pq}^i$ is same as that of $\mathcal{S}_{pq}^i$. An important observation in the modeling presented above is that $\mathcal{B}_{pq}^i$ depends on T_{pq}^i, which in turn depends on $\mathbb{S}_{pq}^i$. Therefore, optimizing T_{pq}^i may seem to optimize $\mathcal{B}_{pq}^i$ as well. However, for a large number of EVs that are required to be scheduled together, the problem of scheduling and routing considered in this paper is not trivial due to the waiting time for the EVs arising out of the fact that the number of CSs is much lesser as compared to the number of EVs in practice.

Considering that an EV i is charged non-preemptively till a threshold battery capacity $\mathcal{B}_h^i$, the total time required by i to travel from the designated source to its destination is:

$$T_{tot}^i = \sum_{\forall t} r_t^i + \sum_{\forall t} \sum_{\forall j \in m} c_t^{ij} + \sum_{\forall t} \sum_{\forall j \in m} w_t^{ij} \tag{6}$$

Since the touring time of an EV depends on the path it takes from source to destination, it can also be represented as:

$$\sum_{\forall t} r_t^i = \sum_{\forall e_{pq} \in E} x_{pq}^i \cdot T_{pq}^i \tag{7}$$

where $x_{pq}^i \in \{0, 1\}$ represents whether the edge e_{pq} is selected to traverse by i for the tour. Replacing the right-hand side of Eq. (7) in Eq. (6), we have:

$$T_{tot}^i = \sum_{\forall e_{pq} \in E} x_{pq}^i \cdot T_{pq}^i + \sum_{\forall t} \sum_{\forall j \in m} c_t^{ij} + \sum_{\forall t} \sum_{\forall j \in m} w_t^{ij} \tag{8}$$

Similarly, the total energy consumed by EV i for the tour can be written as:

$$\mathcal{B}_{tot}^i = \sum_{\forall e_{pq} \in E} x_{pq}^i \cdot \mathcal{B}_{pq}^i \tag{9}$$

It is noteworthy that Eq. (9) does not have the terms c_t^{ij} and w_t^{ij} since we assume that there is no energy consumption while charging and waiting. We further assume that the battery energy capacity of the EVs when reaching their destinations must be within certain limits, as given below:

$$\underline{\mathcal{B}^i} \leq \mathcal{B}_{\rightarrow v_d^i}^i \leq \overline{\mathcal{B}^i} \tag{10}$$

where $\underline{\mathcal{B}^i}$ and $\overline{\mathcal{B}^i}$ are the lower and upper limits on the energy stored in the battery for EV i respectively. Under the above assumptions and settings, the multi-objective optimization problem we are trying to solve

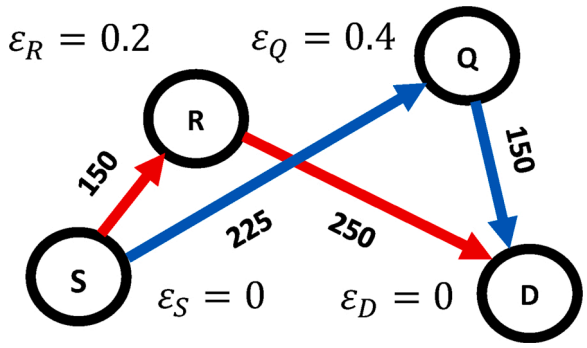


Fig. 2. Example city network with edge weights equal to the actual distances. Nodes have elevation parameters.

here is to minimize the overall touring time and energy consumption for the tour by the whole set of n EVs to their destinations from their respective sources. Mathematically, it can be stated as follows:

$$\begin{cases} \text{Minimize} : \{T_{\text{avg}}, \mathcal{B}_{\text{avg}}\} \\ s.t. (1) - (5), (7) - (10) \end{cases} \tag{11}$$

where $T_{\text{avg}} = \frac{\sum_{\forall i \in N} T^i_{\text{tot}}}{n}$, and $\mathcal{B}_{\text{avg}} = \frac{\sum_{\forall i \in N} \mathcal{B}^i_{\text{tot}}}{n}$.

It is important to note that since the size of the set N is fixed for a particular scenario, the solution to problem (11) will eventually result in optimizing the total time and total energy required for the complete tour for a given set of EVs.

We highlight the significance and complexity of the above multi-objective optimization problem with a simple example. Let us consider two EVs (EV1 and EV2) are to be scheduled and routed through a city network, which can be represented as a graph shown in Fig. 2. The example city network contains four nodes, and the edges are marked with weights representing the distance values. The S and D nodes, both with elevation $\mathcal{E}_S$ and $\mathcal{E}_D$ equal to 0, are assumed to be the sources and destinations for both the EVs, respectively. The network also has nodes Q with elevation $\mathcal{E}_Q = 0.4$, and R with elevation $\mathcal{E}_R = 0.2$. For simplicity, we assume the EVs to share the same configurations viz., maximum speed 80 kmph, power consumption 15 kW, and battery capacity of 60 kWh, which is also assumed to be the threshold battery capacity for both the EVs, i.e., whenever put on to charge, both the EVs are charged till 60 kWh. We consider that both the EVs are fully charged when at the beginning, and they can be charged at a rate of 7kW/h. Since there are two EVs, we assume that EV1 is processed first. As can be observed from Fig. 2, there can be two ways to route the EVs, P1 (marked in blue): $S \rightarrow Q \rightarrow D$ with a total distance of 375 km, and P2 (marked in red): $S \rightarrow R \rightarrow D$ with a total distance of 400 km.

For traditional source-destination routing on plane surfaces, the obvious route would be P1 since it is the shortest one (considering that time is proportional to distance only). However, in this problem set-up, the nodes are associated with elevation values, which affects the speed and eventually the time. For example, with an input speed of 80 km/h the effective speed for EV1 on the edge e_{SQ} is going to be $80/\{1 + (0.4 - 0)\} = 57.1$ km/h, and the effective speed on e_{QD} will be $80/\{1 + (0 - 0.4)\} = 133.3$ km/h. Hence traveling from S to Q will take 3.9 h, and energy consumption of 58.5 kWh, whereas time to travel from Q to D would be 1.1 h, and the energy consumption is 16.5 kWh. Thus, if routed along the path P1, EV1 would spend 5 h in traveling and 75 kWh of battery energy consumption.

Similarly, on path P2, EV1 will require 2.3 h of traveling time and 34.5 kWh of energy consumption on $S \rightarrow R$, and 2.5 h of traveling and 37.5 kWh of energy consumption on $R \rightarrow D$. Therefore, P2 will require a total of 4.8 h of traveling time and 72 kWh of battery energy consumption. Thus, although P1 is shorter in distance, traveling on P2 will require less time and energy for EV1. However, with a battery capacity of 60 kWh, EV1 cannot travel directly from S to D without charging somewhere in between. If P1 is selected, EV1 needs to be charged at Q for 58.5 kWh of energy, which is obtained by subtracting the remaining battery capacity (60–58.5) from the threshold battery capacity (60) and will require 8.3 h of charging time with the 7kW/h charger. Consequently, on P2, EV1 needs to be charged at R for 34.5 kWh of energy, requiring 5 h of charging time. Thus, for EV1, P1 will require a total time of 13.3 h with 75 kWh of energy consumption, and P2 will need a total time of 9.8 h with 72 kWh of energy consumption. Hence, among the two paths, P2 is selected for EV1.

After deciding the route and charging requirements for EV1, EV2 is processed. Since both the EVs have the same configurations, the traveling time and energy consumption on the edges for EV2 is also the same as that of EV1. However, now that EV1 is already scheduled to be charged for 5h at R, traveling along P2 for EV2 would incur a waiting time of 5 h at R (the time for which EV1 is charged). Therefore, the effective total time for EV2 along P2 would be 4.8 (traveling time) +5 (waiting time) +5 (charging time) = 14.8 h. On the other hand, traveling along P1 would eliminate this waiting time for EV2 as Q is free, resulting in a total time requirement of 13.3 h (5 h of traveling and 8.3 h of charging). Thus, for EV2, P1 is better in optimizing time, whereas traveling on P2 would be energy efficient. Thus, the above discussion indicates a trade-off between time and energy consumption, and choosing between P1 and P2 is not trivial as both are Pareto optimal for EV2 in this particular example. Our objective in this work is to obtain such trade-off solutions.

It is important to note that the obtained solutions for the EVs depend on the order in which they are processed. Obtaining the best ordering is computationally intractable, and therefore, we assume that some heuristic orderings of the EVs are already provided to the system as inputs. Due to the intrinsic complexity and hardness of the multi-objective optimization problem presented in Eq. (11), we next propose an efficient and intelligent heuristic charge scheduling and eco-routing for the EVs while satisfying all the conditions imposed by Eqs. (1)–(5), (7)–(10).

3. Proposed multi-objective heuristic algorithm

In this section, we propose a multi-objective heuristic algorithm (MoHA) to schedule the given set of EVs on a city network to minimize average touring time and average energy consumption for the complete set. The proposed algorithm is a graph-based centralized scheduling strategy for the EVs; the pseudo-code of the same is presented in Algorithm 1. The inputs to the algorithm are specifications regarding the city network (such as the number of CSs, the elevation parameters, the types of charging ports in the CSs) and the details of the EVs (such as the number of EVs, their individual maximum, and initial battery capacities, individual sources and destinations, their speed, ranges, energy consumption, charging time, and the ordering in which they are to be processed). The MoHA produces output in the form of a detailed schedule for the EVs from their individual sources to respective destinations while optimizing Eq. (11). The overall functioning of the MoHA can be described in three phases:

Phase #1: **Graph Abstraction and Initialization**: With the given specifications of the CSs, the proposed algorithm constructs a directed graph with nodes representing the CSs, and edges indicating the possible paths among the nodes. Although the actual distances between any two nodes are constant, the actual time required and energy consumed to travel between the nodes vary for each car, depending on multiple factors, including the speed of the vehicle and the slope and length of the path as discussed in Section 2. These values are computed using Eqs. (4) and (5). On this graph, the algorithm schedules each of the given EVs one by one.

Phase #2: **Obtaining Best Path**: Once the edges are assigned weights according to the specifications of the current EV i, the proposed MoHA obtains the route for i from its source to its destination via the best route possible. In this discussion, the best route is the Non-Dominated Shortest Path (NDSP), formally defined as follows:

Definition 1. A path $\mathbb{P}^i_{1\mathbb{SD}}$ for an EV i from source $\mathbb{S}$ to destination $\mathbb{D}$ having a total cost of $W^i_{1\mathbb{SD}} = \langle T^i_{1\mathbb{SD}}, E^i_{1\mathbb{SD}} \rangle$ is said to be a non-dominated shortest path (NDSP) if there exists no other path $\mathbb{P}^i_{2\mathbb{SD}}$ for EV i from $\mathbb{S}$ to $\mathbb{D}$ with cost $W^i_{2\mathbb{SD}} = \langle T^i_{2\mathbb{SD}}, E^i_{2\mathbb{SD}} \rangle$ such that:

1 $T^i_{2\mathbb{SD}} < T^i_{1\mathbb{SD}}$ and $E^i_{2\mathbb{SD}} \leq E^i_{1\mathbb{SD}}$, OR
2 $T^i_{2\mathbb{SD}} \leq T^i_{1\mathbb{SD}}$ and $E^i_{2\mathbb{SD}} < E^i_{1\mathbb{SD}}$, OR
3 $T^i_{2\mathbb{SD}} < T^i_{1\mathbb{SD}}$ and $E^i_{2\mathbb{SD}} < E^i_{1\mathbb{SD}}$

Algorithm 1. Multi-objective heuristic algorithm (MoHA)

Input: The set of m CSs ($\mathbb{C}$), their individual charging port types ($\mathcal{P}_c = \langle \mathcal{P}^1_c, \mathcal{P}^2_c, \ldots, \mathcal{P}^m_c \rangle$), the corresponding elevation parameters ($\mathcal{E}_m$), and the possible paths along with the distances between the nodes. The set of n EVs ($\mathbb{N}$) with their specifications: individual sources ($v_0 = \langle v^1_0, v^2_0, \ldots v^n_0 \rangle$), individual destinations ($v_d = \langle v^1_d, v^2_d, \ldots v^n_d \rangle$), individual energy storage limits ($\overline{\mathcal{B}} = \langle \overline{\mathcal{B}^1}, \overline{\mathcal{B}^2}, \ldots \overline{\mathcal{B}^n} \rangle$) and ($\underline{\mathcal{B}} = \langle \underline{\mathcal{B}^1}, \underline{\mathcal{B}^2}, \ldots \underline{\mathcal{B}^n} \rangle$), individual charging thresholds ($\mathcal{B}_{\text{Th}} = \langle \mathcal{B}^1_{\text{Th}}, \mathcal{B}^2_{\text{Th}}, \ldots, \mathcal{B}^n_{\text{Th}} \rangle$).

Output: The optimized average time requirement (T_{avg}) and the average battery consumption ($\mathcal{B}_{\text{avg}}$) for scheduling and routing all the EVs.

Construct a directed graph $G(V, E)$ having V nodes corresponding to the CSs and E edges connecting the CSs with each other. Each of the nodes has corresponding elevation parameters, and edges are assigned weights equal to the distances between the nodes.
Initialize: $T_{avg} = 0$, $\mathcal{B}_{\text{avg}} = 0$, $T^i_{\text{tot}} = 0$, and $\mathcal{B}^i_{\text{tot}} = 0$.
For each of the EV i in $\mathbb{N}$, **do**:
  For every node $v^i_j (\neq v^i_d)$ (starting from v^i_0) where i halts, **do**:
    Obtain the non-dominated shortest path ($\mathbb{P}^i$) for i from v^i_j to v^i_d.
    Direct i to take the route $\mathbb{P}^i$.
    If $\mathcal{B}^i_{\rightarrow j}$ is sufficient to reach v^i_d, i routed along $\mathbb{P}^i$ till v^i_d without any interruption in between.
      Update $v^i_j = v^i_d$.
    Else halt i at the closest CS v^i_k from v^i_j with the least slack time required (the waiting time + the time required to charge based on the type of charging station) that can be reached with $\mathcal{B}^i_{\rightarrow j}$ along $\mathbb{P}^i$.
      Charge i till $\mathcal{B}^i_h$ at v^i_k.
  End **for**.
  Remove i from $\mathbb{N}$.
  Update T^i_{tot} and $\mathcal{B}^i_{\text{tot}}$.
End **for**.
Return $T_{\text{avg}} = \sum_{\forall i \in n} T^i_{\text{tot}}/n$ and $\mathcal{B}_{\text{avg}} = \sum_{\forall i \in n} \mathcal{B}^i_{\text{tot}}/n$.

From Definition 1, it is quite clear that an NDSP is the one that minimizes both the average touring time and average energy consumption simultaneously and is the best among all the other possible solutions. Therefore, in the current problem set-up, we seek to route each of the EVs along the NDSP from their sources to their destinations.

Starting from the source node, at every iteration, the proposed heuristic mechanism selects an EV i and computes NDSPs from every node at which i halts to charge to its destination node. It maintains a set of halting stations V^i for the EV i. Once the NDSPs for i are obtained, the proposed MoHA directs i to move along that path to reach its destination node v^i_d and follows a recursive method to schedule its charging the CSs in between, if needed. At every node including v^i_0 where i halts, if i has enough energy stored in its battery to be able to reach its destination node from the current node, it is routed through the NDSP from its current node to its destination without any interruption or charging it anywhere in between. However, for cases when the energy level at the current node is insufficient to reach v^i_d directly, the MoHA guides i along its NDSP from the current node but schedules it to halt at the nearest CS from the current node along the NDSP, which has the least slack time (the waiting time + the required charging time at that CS) among the other nodes in that path and can be reached with its current energy level. The aforementioned steps are repeated until i reaches v^i_d. Due to the problem's multi-dimensional nature, there can be multiple NDSPs from a particular node $v^i_j \in V^i$, and theoretically, this number can be exponential. Therefore, to avoid exploring such a large solution space, we consider the following four strategies to select one of the many best solutions:

1 *Time-aware selection*: In this case, the NDSP with the minimum touring time is selected among all the potential NDSPs for an EV.
2 *Energy-aware selection*: On the contrary to the *time-aware selection*, in this case, the NDSP with the least energy consumption is selected among the multiple potential NDSPs for the EVs.
3 *Random selection*: In this strategy, the proposed MoHA selects one of the many possible NDSPs from a particular node randomly.
4 *Weighted selection*: In this strategy, we assign specific weights w_t and w_e for time and energy consumption respectively, and consider the NDSP that minimizes the expression:

$$w_t \times \frac{T^i_{\text{jd}} - T^i_{\text{min}}}{T^i_{\text{max}} - T^i_{\text{min}}} + w_e \times \frac{\mathcal{B}^i_{\text{jd}} - \mathcal{B}^i_{\text{min}}}{\mathcal{B}^i_{\text{max}} - \mathcal{B}^i_{\text{min}}}, \forall i \in N \tag{12}$$

where T^i_{jd} and $\mathcal{B}^i_{\text{jd}}$ are the two cost parameters for the NDSP from the current halting node v^i_j to the destination v^i_d, T^i_{max} and T^i_{min} are the maximum and minimum time on a direct path between two adjacent nodes in the whole network respectively, and similarly, $\mathcal{B}^i_{\text{max}}$ and $\mathcal{B}^i_{\text{min}}$ are the maximum and minimum energy consumption along a direct path between two adjacent nodes in the whole network.

In practice, the values of the above parameters w_t, w_e, T^i_{max}, T^i_{min}, $\mathcal{B}^i_{\text{max}}$, $\mathcal{B}^i_{\text{min}}$ can be input to the system, or may be obtained using Monte Carlo Simulation.

At every iteration, the EV i is either traveling along an edge e_{pq} accumulating the total cost of $W^i_{\text{pq}} = \langle T^i_{\text{pq}}, \mathcal{B}^i_{\text{pq}} \rangle$, or charging, or waiting at some CS accumulating the corresponding charging, or waiting cost (in terms of time).

Phase #3: **Computing the Final Output**: After the completion of phase #2, the overall cost of the tour for all the EVs is computed using Eqs. (8) and (9). The algorithm halts when all the EVs have reached their destinations, returning the average touring time and average energy consumption as output parameters.

In the proposed MoHA, we assume the charging process to be non-preemptive at every CS $v^i_j \in V^i$, i.e., once the charging process starts, it will not be interrupted unless the EV is charged up to a certain threshold

Table 2
EV parameters (Kisacikoglu et al., 2018).

Car model	Battery capacity (kWh)	Max. speed (kmph)	Average energy consumption (kWh) per 100 km	Range (km)	Charging time (min) (with fast charger)
Tesla Model S	70	225	23.6	240	26
BMW i3	22	150	13.5	170	30
Chevrolet Volt	16	160	16.9	61	20
Nissan Leaf S	24	140	15	124	30

Table 3
Electrical ratings of charging methods as per the U.S. standards (Falvo et al., 2014).

Charge method	Nominal voltage (V)	Max. power (kW)
AC level 1	120	1.44
AC level 2	240	7.7
DC level 3	208–600	240

Table 4
Comparative analysis of MoHA with optimal results using the performance metrics for 20 EVs.

Performance metrics	Optimal values for the metrics	energy-aware MoHA					time-aware MoHA					weighted MoHA				
		DBC	*IBC*	*DS*	*IS*	*DE*	*IE*	*DBC*	*IBC*	*DS*	*IS*	*DE*	*IE*	*DBC*	*IBC*	*DS*	*IS*	*DE*	*IE*
ER	0	0.31	0.37	**0.23**	0.36	0.29	0.40	0.30	0.34	**0.20**	0.36	0.27	0.34	0.29	0.33	**0.26**	0.29	**0.26**	0.34
GD	0	0.42	0.45	0.38	0.44	**0.36**	0.42	**0.37**	0.40	**0.37**	0.41	0.40	0.47	0.37	0.43	**0.36**	0.41	0.37	0.40
SS	0	0.67	0.71	**0.47**	0.61	0.59	0.70	0.53	0.67	**0.50**	0.56	**0.50**	0.56	**0.47**	0.63	**0.47**	0.72	0.52	0.58
HVR	1	1.45	1.45	**1.37**	1.43	**1.37**	1.48	**1.41**	1.52	1.45	1.62	1.53	1.58	1.39	1.45	**1.37**	1.57	1.43	1.49

energy level ($\mathcal{B}_h^i$). This consideration is to obtain the waiting time at each of the nodes in a deterministic way. The threshold energy level can be input to the system or may be considered to be $\overline{\mathcal{B}^i}$ for simplicity.

3.1. Timing Analysis of the proposed MoHA

The proposed algorithm iterates through all the EVs, and from every halting CS, it obtains all the NDSPs. The worst-case time complexity for obtaining NDSPs can be $O(|E|)$, and the size of the halting set for an EV can be $|V|$. Therefore, for a single EV, the time complexity to obtain NDSPs from all of its halting CSs is $O(|V||E|)$ at the worst. Along with an NDSP, the algorithm needs to determine the closest node having the least amount of slack time that can be reached with the current energy level for an EV. This process can take $O(|V|)$ time at the worst, resulting in a time complexity of $O(|V||E| + |V|)$ for a single EV. Therefore, for the complete set of n EVs, the worst-case time complexity of the proposed MoHA is $O(n(|V||E| + |V|))$, which is effectively $O(n|V||E|)$ or $O(nm|E|)$ since $|V| = m$.

3.2. Optimality and performance analysis of MoHA

The optimality and performance of the multi-objective optimization problem being discussed depend on two crucial factors: i) the ordering of the EVs in which they are processed, and ii) the optimal paths for each of the EVs on the network. For a set of n EVs, the total number of possible orderings is exponential, which makes the problem difficult to solve for optimal results. Thus, it is NP-hard, and therefore, we consider some heuristic orderings (such as ordering based upon the initial battery capacities or ranges of the EVs) as inputs, making the proposed MoHA intrinsically sub-optimal. Furthermore, since this is a multi-objective optimization problem, the desired solution is the complete Pareto optimal front that includes multiple non-dominated solutions, which is a hard problem. Therefore, the proposed MoHA tries to efficiently obtain one of the Pareto optimal solutions, avoiding a significant part of the complete solution set. Hence, MoHA does not guarantee an optimal solution.

4. Experimental results and discussions

We have implemented the proposed MoHA in Java on a computer having Intel Xeon processor E5-2609 v3 (15 MB Cache, 1.90 GHz, 6 cores) with 132 GB of RAM. We tested the proposed algorithm in a simulated environment on randomly generated synthetic data as well as on a real-life transportation network. To depict a real-life scenario for the experiments, we consider four of the most widely available commercial EVs (Kisacikoglu, Erden, & Erdogan, 2018): Tesla Model S, BMW i3, Chevrolet Volt, and Nissan Leaf S. The specifications of these models are shown in Table 2. Needless to say that the proposed algorithm would also work on any other EV specifications. Table 2 represents the relevant technical specifications for the EVs. The initial battery capacity for all the EVs are assumed to be within [30–60]% of their actual battery capacities, and we assume $\mathcal{E}_{\text{Th}}^i$ to be 80% of the respective battery capacities of the EVs. The sources and destinations for the EVs are randomly assigned to nodes in the graphs, with the obvious constraint of the source and the destination for a particular EV being different in a specific test case. Further, we have considered three types of charging stations, *AC level 1*: the slow charging mode mostly used in residential or office buildings, *AC level 2*: the standard charging mode found in the majority of the public and private spaces, *DC level 3*: the fast charging mode, specifically designed for public usage (Falvo, Sbordone, Bayram, & Devetsikiotis, 2014). The specifications of the various charging levels are shown in Table 3.

As mentioned earlier, the multi-objective optimization problem is studied in this work has two major sub-problems: i) optimal ordering of the EVs and ii) scheduling and routing the EVs on the individual optimal

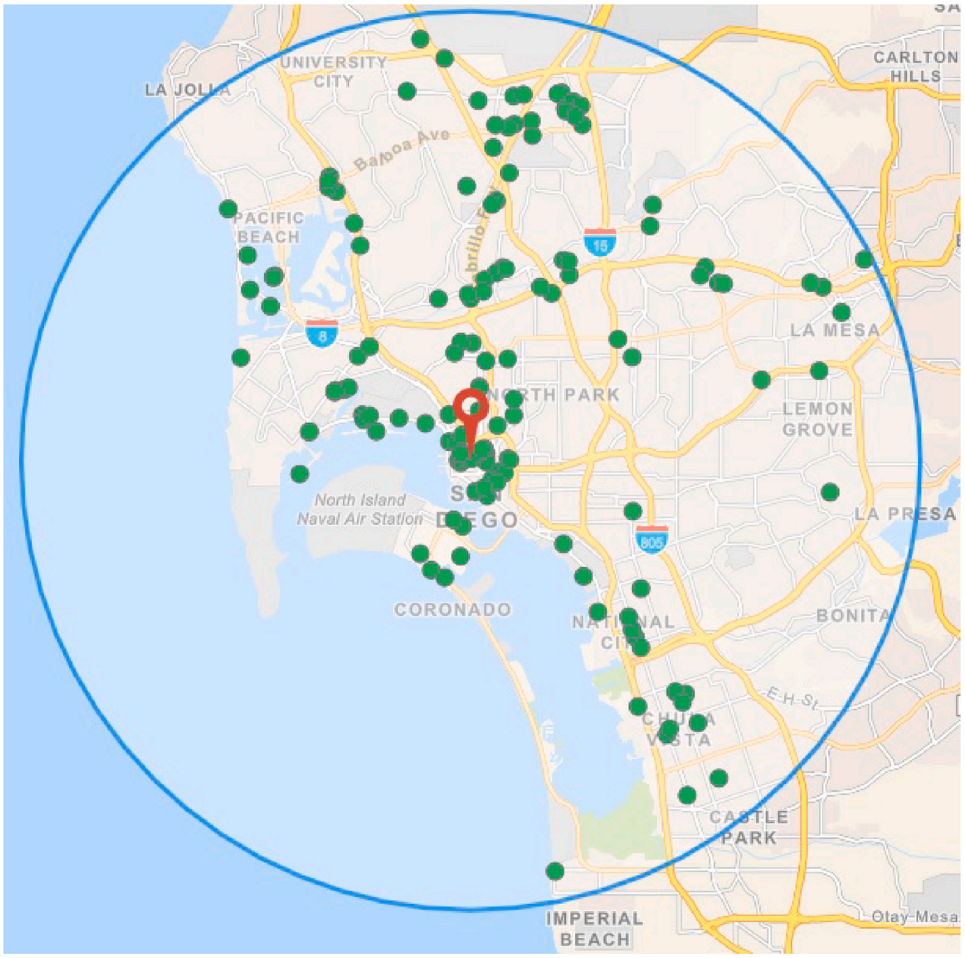


Fig. 3. EV Charging stations in San Diego, U.S. (Alternative, 2020).

paths. Both the sub-problems are intractable, and therefore obtaining the optimal solution for the complete problem is infeasible in a reasonable amount of time. Therefore, we primarily focus on the second sub-problem and consider EVs' ordering statically based on certain properties. In particular, the following six orderings are considered to process the EVs for the second sub-problem: 1) *Decreasing Battery Capacity (DBC)* 2) *Increasing Battery Capacity (IBC)* 3) *Decreasing Speed (DS)* 4) *Increasing Speed (IS)* 5) *Decreasing Energy Consumption (DE)* 6) *Increasing Energy Consumption (IE)*. It is to be noted that there can an exponentially finite number of orderings possible, and we consider only a handful of them to discuss the charge scheduling and eco-routing problem. We use these orderings to evaluate the proposed MoHA. In particular, we evaluated MoHA for up to 10000 EVs with the four NDSP selection strategies as described in Section 3.

4.1. Comparative analysis with the optimal results

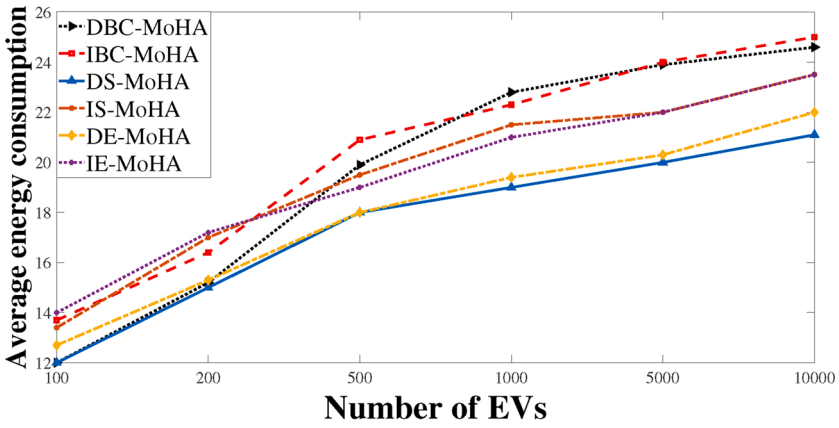
We start the evaluation process by comparing the performance of the heuristics with the optimal solutions. The optimal results are obtained by solving the MILP formulation in IBM ILOG CPLEX v 12.7.1 optimizer on randomly generated graphs of sizes. Being a multi-objective solution to the multi-objective scheduling problem at hand, MoHA results in a collection of vectors instead of a scalar quantity. Therefore, the process of evaluating the solutions and understanding the quality of solutions needs a special set of tools. In this regards, we utilize the available performance evaluation and quality measurements for approximated non-dominated solutions that are widely used in the domain of multi-objective optimization problems. In particular, we use the following quality measures in this paper: *Error Ratio (ER)*[1] Knowles and Corne (2002), *Generational Distance (GD)*[2] Deb (2001), *Schott's Spacing Metric (SS)*[3] Schott (1995), *Hyper-Volume Ratio (HVR)*[4] (Zitzler & Thiele, 1998). The specific reason behind choosing these metrics as performance measures are because of their Pareto compatibility, which tries to minimize the distance of the results obtained from the Pareto optimal set. Thus, lower values for these metrics indicate that the results are closer to the Pareto optimal solutions. Table 4 depicts the results obtained for the variants of the proposed MoHA using these performance metrics, where the best-obtained results are marked in bold for each of the performance metrics. Here we consider discussing the effect of the control parameters and therefore, do not include discussing *random*

[1] ER $= \left(\sum_{i=1}^{n} e_i\right)/n$, where n is the number of vectors in the approximation set and $e_i = 0$ if vector i is in the optimal set, 1 otherwise.

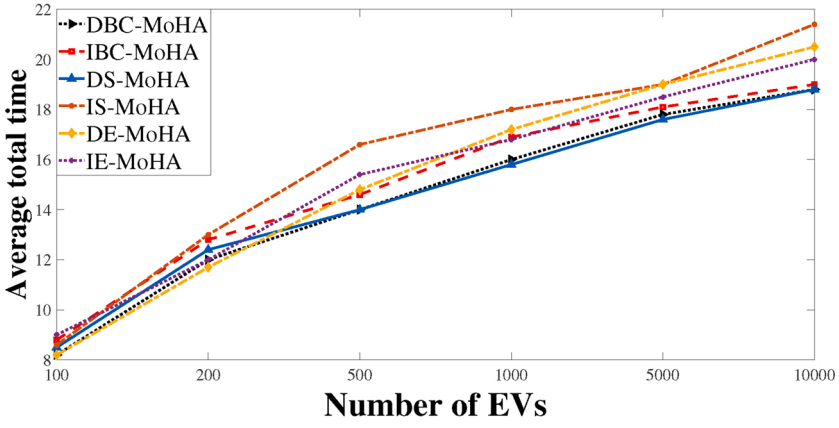
[2] GD $= \left(\sum_{i=1}^{n} d_i^2\right)^{1/2}/n$, where d_i is the distance between vector i and the nearest member of optimal set.

[3] SS $= \sqrt{1/(n-1)\sum_{i=1}^{n}(\bar{d}-d_i)^2}$, where i,j = 1, 2, … n, n being the size of the approximation set, $\bar{d}$ is the mean of all d_i and $d_i = (min_j|f_1^i(\vec{x}) - f_1^j(\vec{x})| + |f_2^i(\vec{x}) - f_2^j(\vec{x})|)$.

[4] HVR $= \frac{H_{obt}}{H_{opt}}$, where H_{obt} is the obtained hyper-volume value by the heuristic and H_{opt} is the hyper-volume value for the optimal points, and $H_x = \bigcup_i a_i|v_i$, where v_i is a non-dominated vector obtained by either the heuristic or optimal method and a_i is the hyper-volume determined by the components of v_i and the origin (0, 0).



(a) Variation of average energy consumption.



(b) Variation of average time requirement.

Fig. 4. Performance evaluation of *energy-aware* MoHA for various pre-orderings.

MoHA. It can be observed from the table that almost all the variants of MoHA have produced results very close to the Pareto optimal front. In particular, the *energy-aware* MoHA and *weighted* MoHA have performance comparatively better than the *time-aware* MoHA. On the other hand, among the ordering heuristics, the DS variant has outperformed the other five variants, except for *energy-aware* MoHA where the DE ordering could also produce better results, specifically for GD and HVR performance metrics. On the other hand, the DBC ordering on *weighted* MoHA performed equally well as compared to its DS counterpart for both *weighted* MoHA and *time-aware* MoHA. It is also noted that the orderings based on the increasing values of the parameters (IBC, IE, IS) could not perform well as compared to their decreasing counterparts.

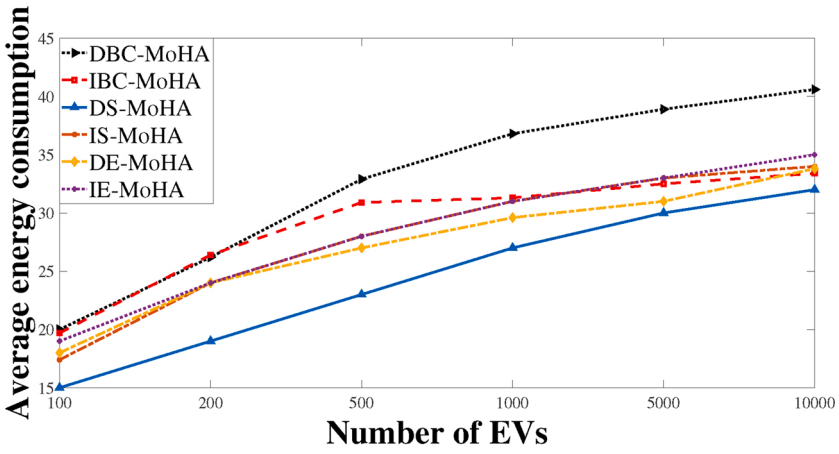
However, a key point here is to note that all the aforementioned performance evaluation metrics need the Pareto optimal front in order to measure the quality of the solutions produced by the proposed MoHA. This requirement makes it difficult to obtain and compare the results for larger test cases due to the intractability of the overall problem. Due to this reason, we evaluate MoHA on larger test cases by performing a comparative analysis among its variants in the next section.

4.2. Analysis on large scale real-world data

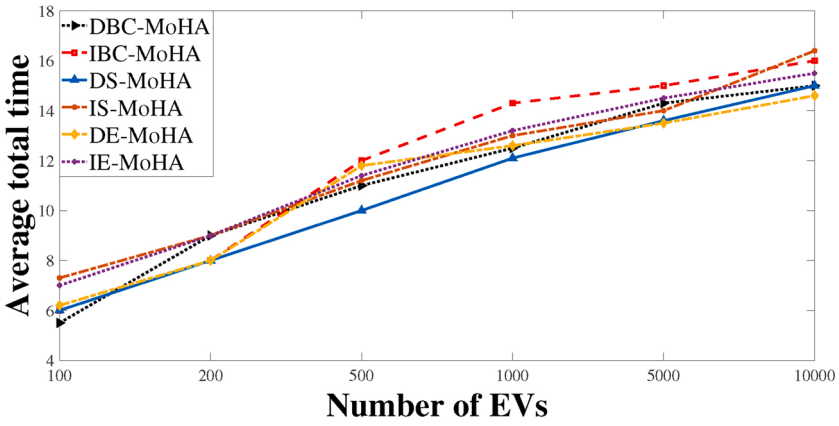
Next, we analyze the performance of MoHA in a simulated environment, depicting a real-world scenario. In particular, we considered the EV charging stations available within 10 miles around the city center of San Diego, the U.S., where there are 150 public EV CSs containing a total of 559 charging outlets (Alternative, 2020). The exact locations and types of the CSs are also available in Alternative (2020). A graph is constructed with these CSs as nodes, which are connected through edges mimicking real roads. Fig. 3 shows a high-level view of the locations of the CSs and the road-map of San Diego (Alternative, 2020). In this set-up, we evaluate MoHA with the four NDSP selection strategies. Figs. 4–7 represent the results obtained for different variants of MoHA for up to 10000 EVs. For each test case size (number of EVs), we experimented with 100 different test cases of same size.

Fig. 4a and b show the performance of *energy-aware*-MoHA in optimizing average energy consumption and average time for the EVs respectively. For *energy-aware* MoHA, the DS and the DE orderings have outperformed the other four orderings in terms of average energy consumption. For a lesser number of EVs, the DBC ordering has also performed well. However, the performance gets degraded with the increase in the number of EVs. On the other hand, for minimizing the total time required for the EVs, all the decreasing orderings (DBC, DS, and DE) have produced competitive results for a smaller number of EVs. However, DS ordering is noted to perform better as the number of EVs grows.

In Figs. 5a and b, the performance of the proposed MoHSA with the *time-aware* strategy is depicted with various initial orderings of the EVs as discussed earlier. Like in the *energy-aware* strategy, the DS ordering has performed significantly better in the *time-aware* MoHA as compared to the other orderings in terms of reducing the average energy consumption of the EVs. The same is shown in Fig. 5a. Overall on average, the DS performed 11.35% better as compared to the other variants. However, there is no single winner for optimizing the average time for this strategy as well. For optimizing the average time requirement, both the DS and DE orderings performed almost equally well. Although one can easily observe from Fig. 5b that the DS ordering has been highly



(a) Variation of average energy consumption



(b) Variation of average time requirement

Fig. 5. Performance evaluation of *time-aware* MoHA for various pre-orderings.

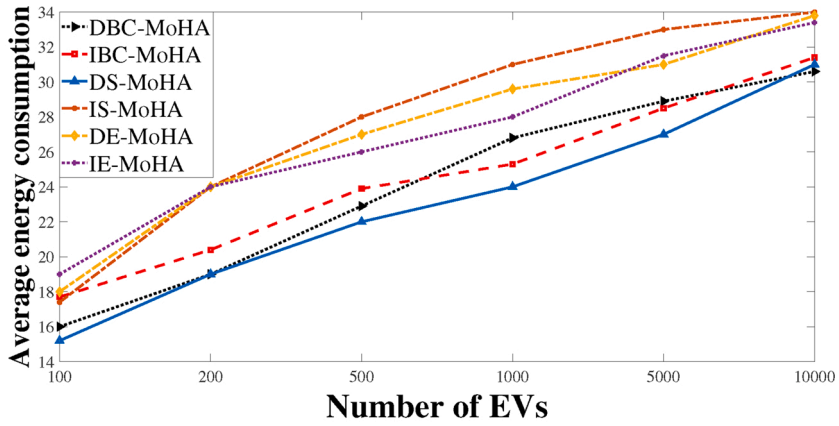
consistent, but the results produced by the DE ordering have been fluctuating.

Results shown in Fig. 6 are due to the *random* MoHA, where the selection of one among the multiple potential NDSPs is random for each of the different test cases for a different number of EVs. Each test case is iterated over 50 times, and Fig. 6 represents the aggregated results. For our implementation purpose, we utilize the *Math.random()* method in Java. As can be observed in Fig. 6a, the performance of the DS ordering on MoHA is still better in optimizing average energy consumption than the other orderings, even though the selection of NDSP is not controlled in this case. However, we do not have any single winner for optimizing the average time requirement, as can be observed in Fig. 6b. Notably, for this *random* strategy, the performance of the DS ordering has degraded when compared with *energy-aware* or *time-aware* strategies in terms of optimizing the average time required for EVs.

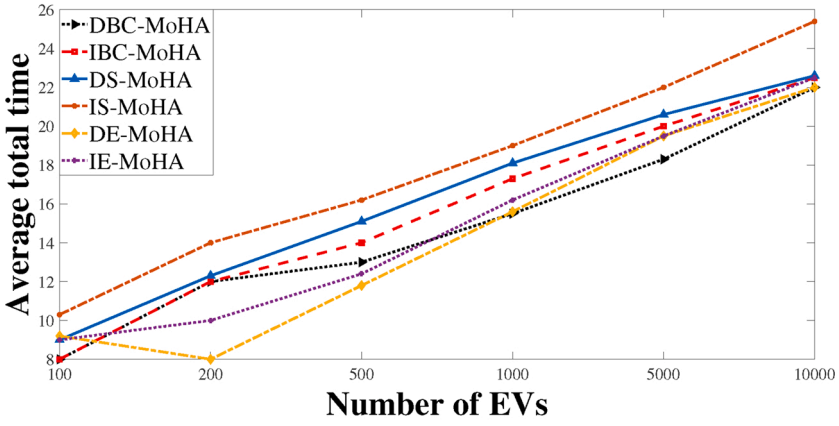
On the other hand, the DE ordering has performed better for a lesser number of EVs, and the DBC ordering has produced better results for a higher number of EVs. Overall, all the orderings, except the IS ordering, have produced similar results in optimizing average time requirements for this *random* NDSP selection strategy in MoHA. Since we do not have any control over this *random* MoHA strategy, we limit discussing this strategy till here only.

Finally, in Fig. 7, we show the results of the *weighted* MoHA, in which we associate certain weights to the cost parameters (energy consumption and time) and choose the NDSP that minimizes Eq. (12). For our experimentation purpose, we have randomly varied the values of w_t and w_e such that $w_t + w_e = 1$. The values of the variables $T^i_{\max}$, $T^i_{\min}$, $\mathcal{B}^i_{\max}$, and $\mathcal{B}^i_{\min}$ are obtained empirically. In Fig. 7, we show the specific scenario where the values of w_t and w_e are considered to be 0.5 each. As can be observed in Fig. 7a, the DS ordering has outperformed all the other orderings once again in minimizing the average energy consumption. In particular, it has performed 9.11% better than the DE ordering, which is the second-best for this weighted MoHA strategy. On the other hand, while minimizing the average, the DBC ordering performed better for cases when the number of EVs is less, while the DS ordering tends to outshine when the number of EVs gets large. The same is being depicted in Fig. 7b. These results are well in sync with what has been observed in Figs. 4 and 5 . It is important to note that the weight combination of $w_t = 1$, $w_e = 0$ is equivalent to the *time-aware* strategy since we consider minimizing time here. Similarly, the weight combination $w_t = 0, w_e = 1$ reflects the *energy-aware* strategy. Therefore, the weight combination of $w_t = 0.5, w_e = 0.5$, the results of which are shown in Fig. 7, reflects the scenario where both time and energy are equally prioritized.

For each of the nodes in the above experiments and simulation re-



(a) Variation of average energy consumption



(b) Variation of average time requirement

Fig. 6. Performance evaluation of *random* MoHA for various pre-orderings.

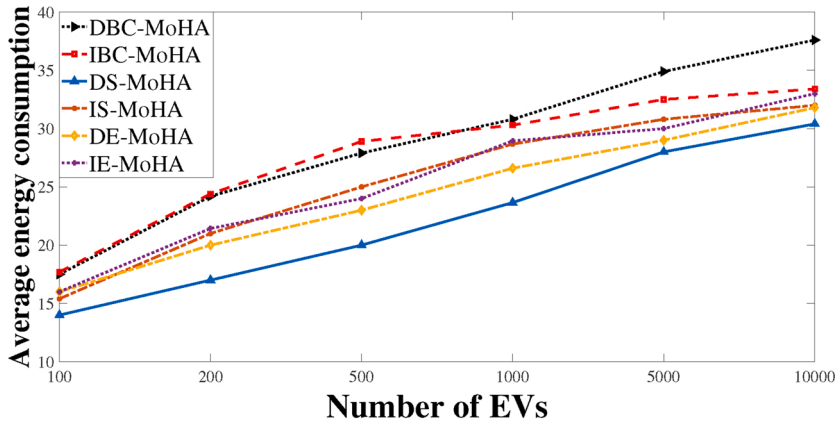
sults, we randomly assigned the charging port type from one of the three types, as shown in Table 3. An interesting scenario that has been observed throughout the experiments is the effect of this charging port type distribution throughout the road network. It has been observed that the performance of *time-aware* MoHA is better for cases when the majority of the CSs have AC level 1 chargers. However, when the number of CSs with DC level 3 chargers increases, *energy-aware* MoHA performs significantly better than the others. Moreover, we have also observed that *weighted* MoHA with weights $w_t = 0.4$ and $w_e = 0.6$ also performs comparatively well with equal distribution of AC level 2 and DC level 3 chargers in the network. Overall, the following key observations can be drawn from the above results shown in Figs. 4–7 and Table 4:

1 For minimizing the average energy consumption, the DS ordering has performed better for the majority of the cases, across all the four NDSP selection strategies of MoHA.
2 There is no single winner for minimizing the average time required for the set of the EVs to complete the tour. However, the two better-performing ordering strategies on this aspect are the DBC and the DS orderings.
3 Comparing among the four NDSP selection strategies, the *energy-aware* strategy of MoHA has produced better results in minimizing the average energy consumption. Similarly, the *time-aware* strategy has performed better among all the other NDSP selection policies in terms of minimizing the average time requirement. This is predominantly due to the fact that both these variants of MoHA prioritize either energy consumption or time while selecting the best routes for the EVs. Specifically, *energy-aware*-MoHA produced 32.39% better results in minimizing energy consumption, and *time-aware*-MoHA performed better in terms of optimizing average time requirements by 24.32%. Thus, both these variants of MoHA produce trade-off solutions between energy consumption and time required to travel for the EVs. On the other hand, the *weighted* strategy produce results in between the two extremes of *energy-aware* and *time-aware* strategies.

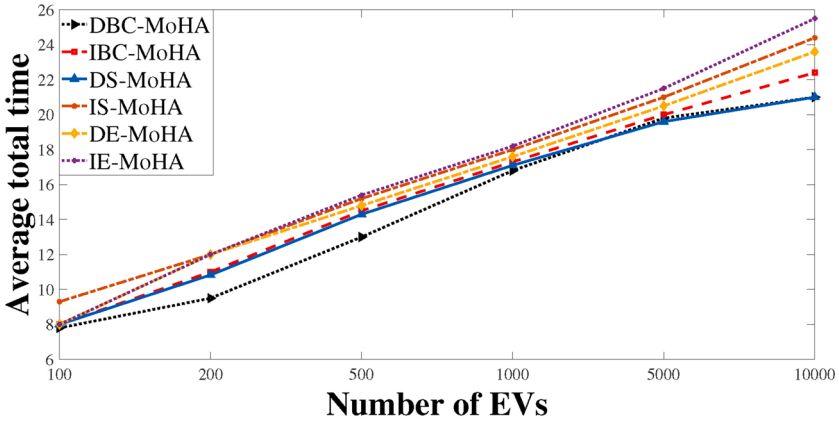
From the above discussions, it is apparent that among the six initial orderings of EVs considered in this paper, the DS ordering has performed comparatively better for most cases, specifically in optimizing the average energy consumption. This observation is consistent irrespective of the four NDSP selection strategies we discussed above. Fig. 8 shows the computation time of the MoHA with DS ordering for the various NDSP selection strategies for different numbers of EVs on graphs of various sizes. As can be observed from the figure, almost for all the NDSP selection strategies, the proposed DS-MoHA produced results in linear time with respect to the input size. However, the *weighted* MoHA consumed a little extra time marginally as compared to the other strategies, which is predominantly due to the additional computation at the run time. Based on this observation, it can be claimed that the proposed MoHA is highly efficient and effective to be implemented in a real-world scenario, with the DS ordering performing better in terms of optimizing the total energy consumed for the given set of EVs.

5. Conclusions and future work

In this paper, we discuss the multi-objective optimization problem of



(a) Variation of average energy consumption



(b) Variation of average time requirement

Fig. 7. Performance evaluation of *weighted* MoHA for various pre-orderings.

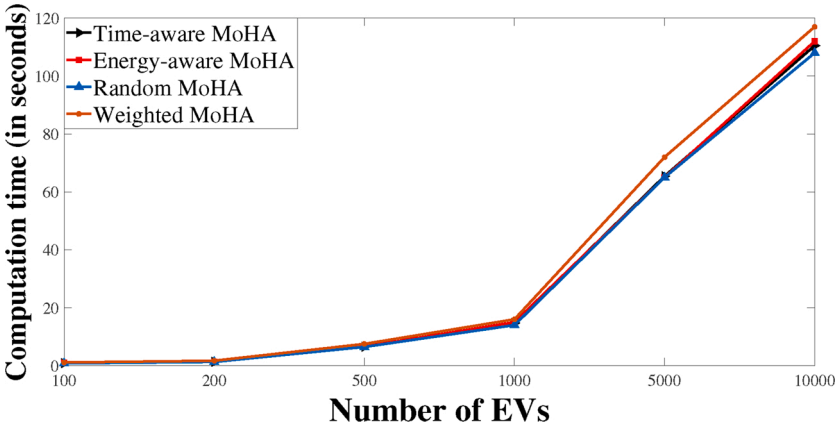


Fig. 8. Computation time for the DS ordering for various NDSP selection strategies for MoHA.

scheduling and routing of EVs to minimize both time and energy consumption. The problem is composed of two sub-problems, which are themselves intractable, making the whole problem hard to solve optimally in polynomial time. We presented a detailed MILP formulation of the problem and proposed an efficient multi-objective heuristic algorithm (MoHA) and its variants to obtain desirable solutions efficiently. We discussed the time complexity of the proposed mechanism and evaluated its performance experimentally on practical data sets. The key findings from the experimental results are:

1 When compared to the optimal solutions using the various performance metrics for multi-objective solutions, the *energy-aware* MoHA and the *weighted* MoHA performed better than the *time-aware* MoHA. On real-world data, among the various NDPS selection strategies, *energy-aware*-MoHA produced 32.39% better results in minimizing energy consumption and *time-aware*-MoHA performed better in terms of optimizing average time requirements by 24.32%.
2 As compared to all the orderings for the EVs discussed in this paper, the DS ordering of EVs yielded better results in reducing energy

consumption and travel time for the EVs to complete their journeys. This is true for all the NDSP selection strategies presented here.

Using the results, we could show that there is a trade-off between energy saving and time-saving. Among the two optimization criteria, preference should be given to energy minimization to obtain better solutions. We have also observed that with variations in the distribution among the types of charging ports, the performance of the proposed algorithm varies. As part of our future work, we are looking forward to developing a more practical and probabilistic model of the problem whereby realistic road and traffic conditions would be incorporated into the modeling. Moreover, obtaining an online variant of the proposed mechanism is a part of our on-going work.

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